

Project Report on

Development of a Digital Twin-Based Predictive Maintenance System for Industrial Motor

BACHELOR OF TECHNOLOGY

in

INTERNET OF THINGS

by

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Internet of Things (IoT)

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CERTIFICATE

This is to certify that the project entitled “**Development of a Digital Twin-Based Predictive Maintenance System for Industrial Motor**” is a bonafide work of **Keyur Chauhan (04), Shivangi Chouhan (06) and Arshiya Khan (14)** submitted to the Thakur College of Engineering and Technology, Mumbai (An Autonomous College affiliated to University of Mumbai) in partial fulfillment of the requirement for the award of the degree of “**Bachelor Of Technology**” in “**Internet of Things**”.

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PROJECT APPROVAL CERTIFICATE

This project report entitled “*Development of a Digital Twin-Based Predictive Maintenance System for Industrial Motor*” by *Keyur Chauhan (04), Shivangi Chouhan (06) and Arshiya Khan (14)* is approved for the degree of “**Bachelor Of Technology**” in “**Internet of Things (IoT)**”.

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ABSTRACT

Industrial induction motors are critical components in manufacturing and automation systems, where unexpected failures can lead to significant downtime, financial losses, and safety risks. Traditional maintenance approaches, such as reactive and preventive maintenance, lack real-time insight into equipment health and often result in inefficient operations. This project presents a Digital Twin-based Predictive Maintenance System for industrial motors, integrating real-time sensor data, machine learning, and interactive visualization.

The system utilizes an ESP32 microcontroller interfaced with MPU6050 (vibration) and DHT11 (temperature and humidity) sensors to continuously acquire motor condition data. Since real-world fault data is limited, a Digital Twin-driven data generation approach is employed to simulate realistic fault conditions such as bearing defects, stator faults, rotor faults, and overheating. This enables the creation of a comprehensive dataset for training and validation.

A multi-layer machine learning architecture is implemented to ensure robust fault detection and classification. The first layer uses Isolation Forest for anomaly detection based on healthy motor behavior. The second layer incorporates physics-informed rule-based logic to identify strong fault signatures using domain knowledge. The final layer employs an XGBoost classifier, trained on approximately 7,750 data samples, achieving an overall accuracy of around 99% in fault classification. Additionally, the system computes a health score (0–100) and estimates Remaining Useful Life (RUL) to provide actionable maintenance insights.

A real-time web-based dashboard using React and Three.js visualizes the digital twin of the motor. The 3D model allows user interaction such as rotation, zoom, component inspection, and X-ray view of internal parts. Sensor data is streamed via WebSocket, dynamically updating motor conditions, graphs, fault status, and maintenance recommendations.

The proposed system successfully demonstrates accurate fault detection, real-time monitoring, and intuitive visualization, providing a cost-effective and scalable solution for predictive maintenance in industrial environments. Compared to traditional systems, it offers improved reliability, early fault detection, and enhanced decision-making capabilities, making it suitable for modern Industry 4.0 applications.

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Chapter 1

Overview

1.1 Introduction

Industrial induction motors are frequently employed within the realm of industrial processes, performing such functions as pumping, compression, and transportation through the use of conveyor belts. Induction motors undergo continuous use within challenging conditions, which causes the emergence of such motor-related faults as heating, vibrations, and mechanical faults.

Reactive and preventive maintenance techniques do not provide efficient solutions to solve such challenges. On the one hand, the former type of maintenance leads to unplanned stoppage of machines, whereas the latter leads to unnecessary maintenance procedures. Therefore, there is a need for intelligent systems aimed at equipment condition monitoring and failure predictions.

This project aims to develop a digital twin-based predictive maintenance system.

1.2 Background

In the age of Industry 4.0, numerous new methods and concepts were created, which made monitoring systems intelligent and connected. Technologies, such as Industrial IoT, machine learning, and digital twins became key factors in developing systems capable of real-time analysis.

Digital Twin is defined as a virtual model of any physical object that reflects changes happening to the latter in real-time thanks to continuous data input from sensors. For this particular project, the data about vibration, temperature, and humidity are collected using ESP32 microcontrollers, processed using machine learning algorithms.

Because there was no access to real-world fault data, a data generation technique was developed following the principles of Digital Twins, which would create a realistic environment for motor fault detection.

1.3 Importance of the Project

As a consequence of faults in a motor working in industrial conditions, such negative effects as losses in production, additional maintenance expenses, and potential safety issues may arise. That is why early fault detection is essential.

The following advantages of the presented project must be highlighted:

- Provides for real-time condition monitoring of the motor
- Makes it possible to detect faults early due to machine learning techniques
- Minimizes unplanned downtimes and additional expenses
- Includes a digital twin visualization interface
- Promotes predictive maintenance strategy

1.4 Perspective of Stakeholders and Customers

The following stakeholders and customers will benefit from the proposed system:

- Maintenance engineers – receive early alerts and fault detection details
- Industry operators – increase efficiency of the system operation
- Management – achieve cost savings, efficient maintenance management
- Technicians – get access to the digital twin visualization interface for diagnosis purposes
- End users/customers – achieve efficient maintenance services and reliable operations

1.5 Objectives and Scope of the Project

Objectives:

- To develop a Digital Twin-based predictive maintenance system
- To collect real-time sensor data using ESP32 and IoT sensors
- To implement machine learning models for fault detection
- To detect anomalies and classify faults in motor operation
- To compute health score and estimate Remaining Useful Life (RUL)
- To design an interactive 3D dashboard for visualization

Scope:

- Focuses on industrial induction motors
- Uses vibration and temperature data for monitoring
- Detects faults such as bearing, stator, rotor, and overheating
- Provides real-time visualization and analytics
- Can be extended to other industrial equipment

1.6 Summary

This chapter gave an overview of industrial motor systems and discussed the shortcomings of conventional maintenance strategies. The concept of predictive maintenance through Digital Twin technology was introduced, along with the relevance of real-time monitoring and intelligent

fault detection. The project's objectives and scope were laid out clearly, defining the core functionalities and boundaries of the proposed system. Altogether, this chapter builds the groundwork for understanding why such a solution is necessary and what it intends to achieve.

Chapter 2

Literature Survey

2.1 Introduction

With Industry 4.0 reshaping the way industrial systems are managed, predictive maintenance has grown into a critical strategy for improving equipment reliability and operational efficiency. Technologies like the Internet of Things (IoT), machine learning, and Digital Twin (DT) systems now make it possible to continuously monitor and assess the condition of machinery.

A Digital Twin essentially creates a virtual counterpart of a physical system, keeping both in sync in real time. When this is paired with live sensor feeds and machine learning, it becomes a powerful tool for early fault detection and predictive analysis. This chapter reviews existing research on Digital Twin systems, IoT-based monitoring approaches, and machine learning techniques applied to predictive maintenance.

2.2 Literature Survey

Quite a few studies have examined how Digital Twin technology and machine learning can be applied to industrial and smart monitoring systems.

Song et al. (2024) proposed a data-driven Digital Twin system for predicting battery states, using machine learning models such as LSTM and DNN. The system places strong emphasis on explainability, incorporating techniques like SHAP and LIME to keep predictions interpretable in safety-critical settings. That said, the work is designed specifically around electrochemical battery behavior and does not address the vibration characteristics or electromechanical dynamics that define industrial motor environments.

Sharaf and Elgazzar (2024) published a broad survey on Digital Twins in IoT-based energy systems, covering areas like sensor integration, communication protocols, and modeling strategies. The paper lays out a solid theoretical framework for energy optimization in smart grids. While it is thorough in scope, the work stays largely conceptual and does not provide the kind of practical hardware-level guidance needed for real-time, edge-based motor monitoring deployments.

Dos Santos et al. (2023) built a Digital Twin monitoring system for induction motors, combining IoT sensors with thermo-magnetic finite element analysis (FEA). The system performs well in mapping temperature and magnetic flux, but the reliance on computationally heavy simulations makes it difficult to run on low-cost microcontrollers, limiting its suitability for decentralized or resource-constrained industrial setups.

Rassölkin et al. (2022) developed an interface for an electric motor Digital Twin using empirical performance models to enable real-time visualization. The system effectively bridges physical sensor readings and a virtual dashboard for monitoring motor behavior. However, the platform does not include automated predictive analytics or ML-based fault detection, meaning maintenance decisions still rely on manual data interpretation.

Cabral et al. (2024) implemented a Digital Twin for a robotic manufacturing cell, following the ISO 23247 standard and using MQTT for data transmission alongside Node-RED for visualization. The study demonstrates that open-source tools can be quite effective in industrial automation contexts. The focus, though, remains on motion mirroring and process geometry rather than the predictive health monitoring of the motors powering the robotic components.

Hassija et al. (2025) carried out a high-level survey on technologies enabling Digital Twins, including 6G communications, AI, and Big Data analytics, across sectors ranging from healthcare to construction. The paper serves as a useful reference for future technology integration. However, it does not go deep into the specific preprocessing steps for multi-sensor vibration and current data, which are particularly important for induction motor health monitoring.

Sun et al. (2021) introduced a physics-informed Digital Twin for electric motor condition monitoring, combining data-driven methods with the physical laws governing motor dynamics. This hybrid approach improves prediction accuracy, especially in unusual or edge-case scenarios. Still, incorporating real-time physical equations adds considerable computational overhead, which can compromise system responsiveness in time-sensitive industrial settings.

Zhang et al. (2023) presented a Digital Twin-based framework for detecting electrical faults in induction motors online, targeting stator winding failures and broken rotor bars. The approach uses intelligent data fusion to differentiate between fault signatures. While it handles electrical anomalies well, the system does not cover mechanical issues such as bearing degradation or shaft misalignment as thoroughly, which limits its applicability in general motor maintenance contexts.

Xu et al. (2022) investigated data-driven prognosis for rotating machines using Digital Twins paired with deep learning. The research demonstrates that convolutional neural networks are effective at extracting meaningful features from raw sensor signals. However, the validation was conducted mostly on controlled laboratory data, which raises questions about how well the system would perform in real industrial environments where noise levels and load conditions vary considerably.

Adamou et al. (2024) proposed a high-fidelity Digital Twin modeling approach for squirrel cage induction motors, specifically aimed at diagnosing broken rotor bars. The study highlights how accurate virtual modeling can help reflect physical degradation. The limitation, however, is that it targets a single fault type and does not offer a generalized interface for monitoring multiple health parameters simultaneously.

The major limitations identified across existing systems include:

- Lack of integration between real-time sensor data and predictive machine learning models
- Limited use of interactive 3D visualization or AR-based Digital Twin interfaces
- High computational complexity due to FEM-based simulations
- Lack of lightweight and cost-effective solutions suitable for edge devices
- Absence of systems combining prediction, visualization, and real-time monitoring

These gaps collectively point toward a clear need for an integrated system that brings together real-time data acquisition, ML-based fault prediction, and interactive Digital Twin visualization in a way that is both cost-effective and scalable.

2.3 Problem Statement

Industrial induction motors form the backbone of many manufacturing operations and typically run continuously under varying load and environmental conditions. When these motors fail unexpectedly, the consequences range from costly production downtime to serious safety hazards.

Conventional maintenance practices, whether reactive or schedule-based, offer no real-time visibility into motor health. Many existing Digital Twin platforms focus mainly on visualization

without incorporating intelligent prediction, while ML-based systems often lack real-time data integration and accessible user interfaces.

What is needed is a unified framework that combines live sensor data, predictive analytics, and interactive visualization within a single system. This project sets out to build exactly that: a Digital Twin-based Predictive Maintenance System that uses IoT sensors, machine learning models, and a 3D interactive dashboard to monitor motor health, identify faults early, and deliver actionable maintenance recommendations.

2.4 Summary

This chapter reviewed a range of existing research covering Digital Twin technology, IoT-based monitoring, and machine learning approaches for predictive maintenance. Several recurring limitations were identified, including insufficient real-time integration, the absence of immersive or interactive visualization tools, and the computational overhead associated with simulation-heavy systems. These observations establish a clear motivation for developing an integrated, lightweight, and real-time predictive maintenance solution, which forms the basis of the work presented in subsequent chapters.

Chapter 3

Planning & Design

3.1 Introduction

In this chapter, the design and planning of the proposed Digital Twin-based Predictive Maintenance system will be described. The system architecture, methodology, and various tools used in the development will be introduced along with some considerations during design.

3.2 Project Planning

For creating the digital twin of a real object, the development of the system requires a proper integration of both hardware components and software modules for delivering an efficient product. As for the hardware side, the first component is the ESP32 microcontroller which acts as the main source of acquiring data from sensors. For the monitoring of vibrations, MPU6050 sensor is connected to it which is then combined with DHT11 for capturing temperature and humidity information. This entire hardware setup will be used on the real industrial RAMAN motor setup. As far as the software part of the system is considered, Python programming language is chosen as the tool for developing machine learning models due to the availability of various libraries. On the other hand, Flask is selected for handling backend tasks and providing an API for communication. For visualization purposes, React along with Three.js is preferred to be used for building interactive 3D dashboards. Unity3D is selected as one of the reference applications to create a model of the digital twin. For simulating and generating training data, MATLAB has been used. MQTT and HTTP are selected for the communication protocol among components. The choice of these tools is based on their scalability, modularity, and cost-effective nature.

3.3 Scheduling

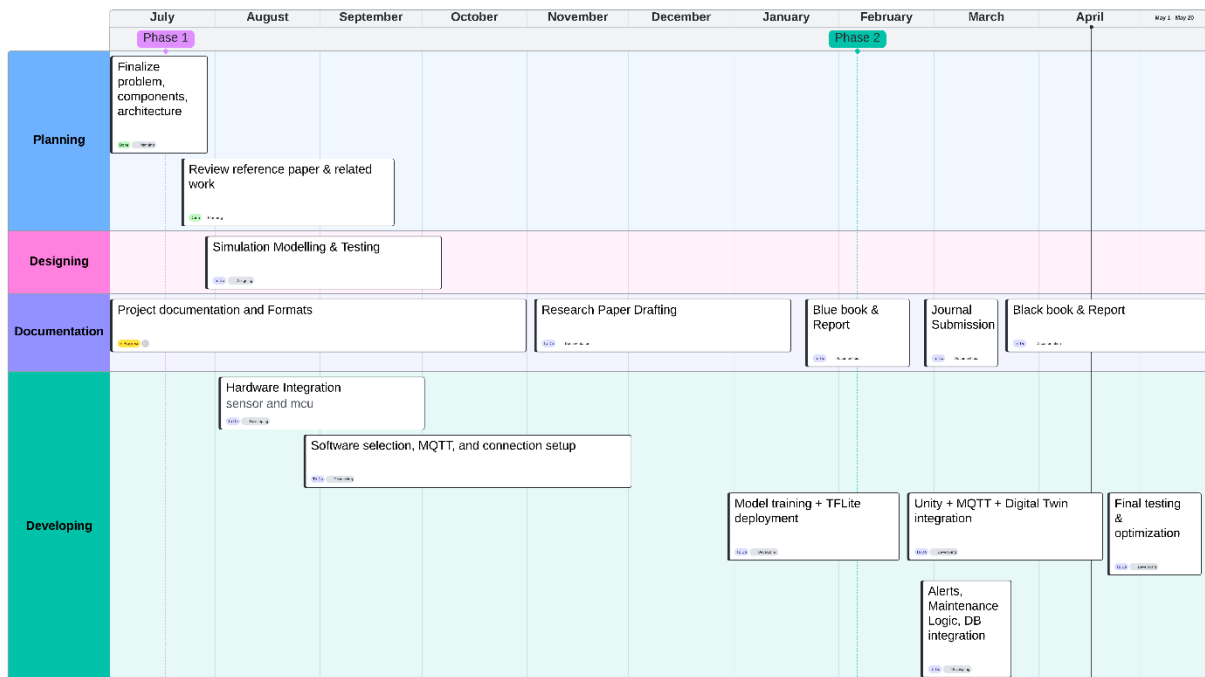


Figure 3.1: Gantt Chart of Project Timeline

The project was conducted in phases within a span of ten months following a systematic approach. This process was initiated with requirements analysis and literature survey in order to create a robust theoretical base, and followed up with designing and simulation of the system architecture. Documentation for this project was generated systematically during all the stages of the project, ranging from the creation of initial drafts to final reports and research submission. Development of the system was done in two phases, with phase one concentrating on hardware set-up and initial level of integration and the phase two involving more advanced activities like machine learning model creation, backend integration, and visualization. In the final phase, the testing, validation, and optimization of the system were completed in order to ensure that it performed reliably. As shown in the figure below (Figure 3.1), there was an equal amount of activities done in planning, designing, developing, and testing phases in this project.

3.4 Proposed System

The system proposed will consist of layers that acquire live data, perform intelligent analysis, and display a digital twin. Its core idea lies in detecting and predicting faults in industrial electric motors using sensors and intelligent algorithms.

3.4.1 Feasibility Study

Technical feasibility lies in the usage of easily obtainable sensors, while economic feasibility comes from the cost-efficiency of components used. For example, the ESP32 used in this project is economically viable to manufacture, and can be acquired relatively cheaply. Operational feasibility lies in the user-friendly dashboard where the results of intelligent analysis are presented in an understandable format.

3.4.2 Block Diagram

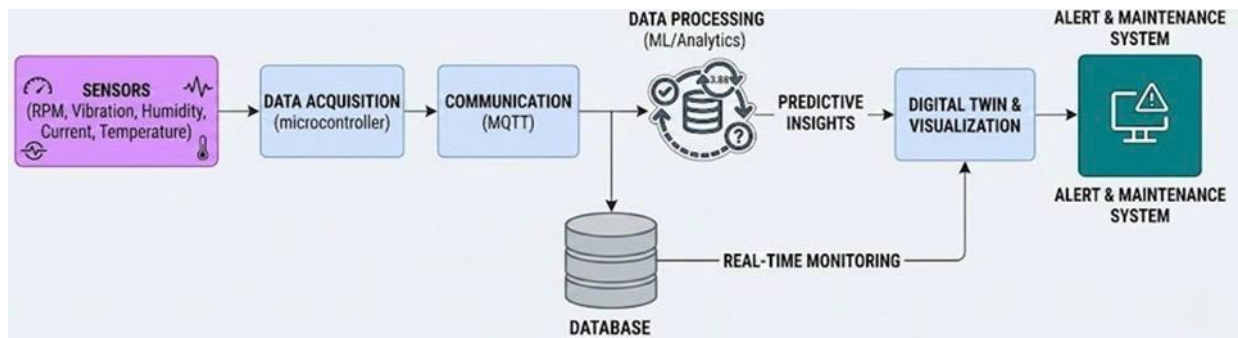


Figure 3.2 Block Diagram of Proposed System

The first step in the operational workflow involves the use of a set of sensors that collect essential information concerning the motor's operational status such as RPM, vibration, humidity, current, and temperature. The collected raw data is then transmitted from the microcontroller via the MQTT protocol. Once the data has been successfully transmitted, it is sent to the database for historical storage and also to the processing unit for further processing. The processing unit employs machine learning models and analytical methods to mine valuable insights from the collected data. Consequently, this architecture facilitates both real-time monitoring and data analysis. Finally, the output of the processing unit is fed into the visualization interface where they are represented in an interactive visual form, thus forming the digital twin.

In addition, the visualization interface displays changes in system behavior in order to provide useful feedback on changes in the performance parameter values. Further, the insights generated from the processed data are fed into the alerts and maintenance unit to monitor and detect any faults and generate warnings about the same.

3.4.3 Methodology

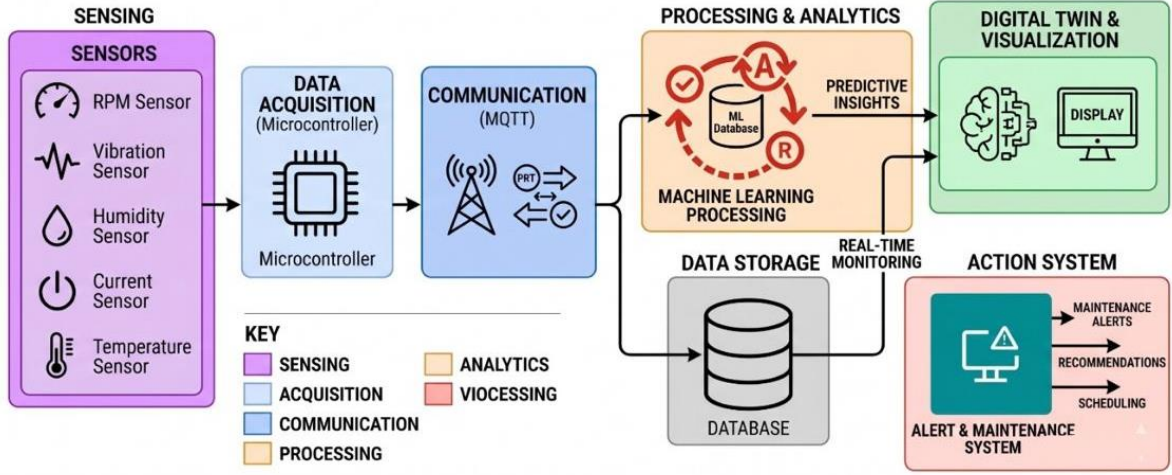


Figure 3.3: System Architecture of the Proposed Predictive Maintenance System

Figure 3.3 illustrates the workflow of the entire system. Data acquisition is done using sensors attached to the motor that collects relevant data such as vibration, temperature, humidity, and current. The collected data is then acquired by the microcontroller at set time intervals and subsequently transmitted to the backend using HTTP/MQTT protocol.

Once the data is received by the backend, it can follow either one of the two parallel pathways. First, the data is stored in the database and made available for real-time monitoring and historical analysis. Second, the data undergoes preprocessing and further processing before being fed into the machine learning model for fault detection.

Finally, the processed data outputs can be fed into the digital twin interface where an interactive visualization will represent the current state of the motor. Additionally, using synthetic data obtained from a MATLAB simulation model, faults are detected despite the absence of actual fault data.

3.4.4 Framework/Algorithm and /or Design details

The development framework follows a data-driven and layered ML approach that converts data coming from the sensors into meaningful insights. The first step involves the collection of data using ESP32 before carrying out preprocessing and feature extraction by way of the sliding window method. Among others, RMS, standard deviation, kurtosis, skewness, and temperature trend were among the statistical features calculated.

In a bid to counter the challenge of inadequate fault data, the proposed system uses a combination of real and synthetic datasets for training. Besides real and synthetic data, a free-to-use database from Kaggle was also included. This open source database includes data on the motor fault conditions which enabled the validation of the performance of the models in the machine learning architecture.

A multi-layered machine learning framework will be deployed in the development. The first layer is made up of an isolation forest anomaly detection algorithm. Through the first layer, the system learns normal operation and uses it in identifying anomalies. Next, a rule-based layer is used in determining strong faults. The third layer is an XGBoost multi-class classifier.

As far as the back-end system is concerned, it has been developed using Python language with Flask framework. The processing is done in real-time basis. At the output stage, information such as fault type, health scores, and RUL values will be provided. This output information will be transmitted to the digital twin dashboard and displayed via graphs and 3D models.

3.5 Summary

This chapter discussed the planning and design of the proposed Digital Twin-based Predictive Maintenance System. The chapter has presented the framework that includes both hardware and software. It also explained the use of a data-driven, multi-layered machine learning system to achieve real-time monitoring and fault detection purposes.

Chapter 4

Implementation & Experimental Set up

4.1 Introduction

This chapter highlights the implementation approach and experiment configuration of the suggested predictive maintenance system using digital twins. It discusses how the theoretical design from Chapter 4 is implemented into a practical system by integrating various hardware and software tools as well as applying machine learning techniques.

The key aspect of the system's implementation is the creation of an efficient pipeline for acquiring raw sensor data, processing it through an intelligent backend, and displaying results on the digital twin dashboard. Apart from the data acquired from real sensors, the system makes use of artificial data generated to simulate faulty behavior.

4.2 Software and Hardware Set up

Component	Description
ESP32 Microcontroller	Used for real-time data acquisition and communication
MPU6050 Sensor	Captures vibration data along X, Y, and Z axes
DHT11 Sensor	Measures temperature and humidity near motor casing
Sensor Interface	Connected via GPIO pins (I2C for MPU6050)
Data Collection	Sensor readings taken at ~500 ms intervals
Motor Setup	Complete system mounted on RAMAN motor
Data Transmission	ESP32 sends data to backend using HTTP (JSON format)

Table 4.1: Hardware Setup Details

Component	Description
Python Backend	Handles preprocessing, feature extraction, and ML prediction
Flask API	Provides endpoints (/ingest, /predict) for data communication
ML Models	Isolation Forest, Rule-based logic, and XGBoost used for prediction

Feature Processing	Sliding window used for statistical feature extraction
MATLAB Digital Twin	Generates synthetic fault data for training
Dashboard (React + Three.js)	Displays real-time system status
Visualization Features	3D motor model, graphs, health score, fault status, recommendations

Table 4.2: Software Setup Details

This system works in real-time, whereby ESP32 sends sensor data to the backend, the backend performs computation through machine learning models, and the outcome is visualized through the dashboard.

4.3 Performance Evaluation Parameters

To assess the effectiveness of the system, several performance parameters are measured for the purpose of capturing analytical accuracy and real-time responsiveness. One of the performance parameters includes classification accuracy that captures how effective the machine learning model is at classifying different fault cases. Moreover, the ability of the proposed system to detect faults is also measured depending on the ability of the system to differentiate different faults such as bearing failures, stator imbalance, rotor failure, and overheating.

A health score is computed and expressed on a scale of 0 to 100 to capture the health status of the motor system. This score depends on vibration and temperature features and is a continuous measurement tool compared to classification. On the other hand, the Remaining Useful Life (RUL) metric is used to estimate how many days the motor will operate without fault.

Finally, real-time response is considered as a performance parameter to evaluate the effectiveness of processing data and updating dashboards within a short period. Consequently, these parameters enable us to capture the efficiency, accuracy, and practicality of the system.

4.4 Implementation and Testing of Modules.

The system is built out of modules that perform specific functions in the pipeline of the workflow.

The data acquisition module continually collects data about motor status from sensors using ESP32 at regular intervals. The data consists of vibrations measured in multiple directions and temperature readings. The collected data is sent to the backend server for further analysis.

When the data is received, the data processing module cleanses the dataset from noise and handles missing values. Next, it groups consecutive sensor readings using sliding windows and calculates their statistical properties, including RMS value, standard deviation, kurtosis, and temperature trends. This step transforms the raw sensor data into more informative data points that can be analyzed over time.

To overcome the lack of actual fault data, a special data generation module was introduced. Based on the digital twin approach, the model simulates motor behavior under various fault conditions and produces realistic data sets that can be used for machine learning purposes.

The machine learning module is a three-layered system, which increases its robustness. First, Isolation Forest detects anomalies and learns what normal operation looks like. In the second layer, the system uses physics-informed thresholds for identifying fault situations and classifying them with a certain degree of confidence. Finally, when patterns overlap and it is difficult to determine the fault type, XGBoost classifier is employed.

The backend integration module is built using Flask framework and acts as an interface for exchanging information between software and hardware components. The module accepts data, executes predictive maintenance tasks, and provides access to the API endpoints.

The visualization module is a user-friendly interface where all output data is displayed. The dashboard provides up-to-date sensor data, trend graphs, classifications of faults, and metrics related to motor health. In addition, there is a 3D motor model on the dashboard that shows how the motor is performing at any given moment.

4.5 Deployment

The system is implemented locally with all the hardware, software, and interface tools embedded in one local platform. In this context, the ESP32 microcontroller sends sensor data to the backend server through HTTP request, and machine learning models process the data. The backend produces predictions and presents the results to the dashboard through API endpoints, whereby the dashboard continuously retrieves data and updates visualization. Consequently, the implementation guarantees seamless interactions among different parts of the system.

4.6 Screenshots of Circuits/GUI/Structure

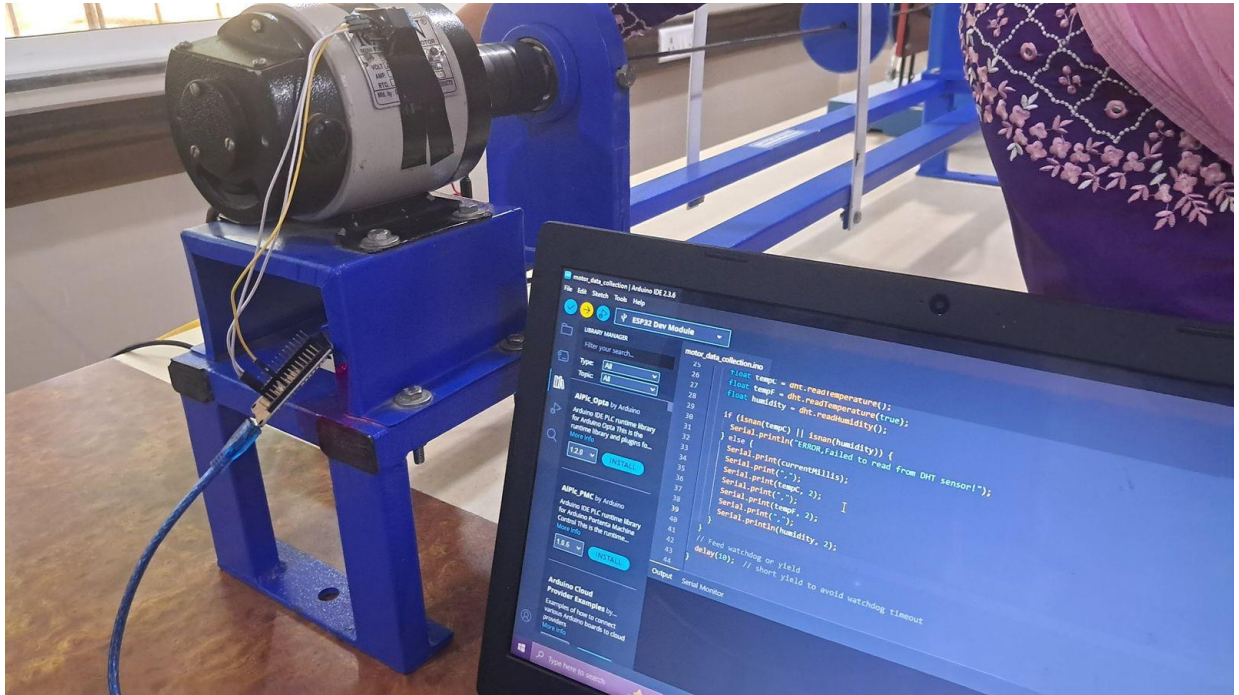


Figure 4.6: Experimental Setup of Motor with ESP32 and Sensor Interface

The diagram displays the experimental setup of the proposed system, where a RAMAN industrial motor is embedded in the ESP32 microcontroller for real-time data acquisition. This is accomplished by connecting sensors to the microcontroller to acquire data on parameters such as vibrations and temperatures. The ESP32 is interfaced with the computer through USB cable where the sensor data is monitored and processed using Arduino IDE.

4.7 Summary

This chapter gave details about the implementation and experimental setup for the proposed system. It explained how sensor data, hardware components, the software frameworks, and machine learning models are integrated with each other to form a real-time predictive maintenance system. The chapter also highlighted the modular design of the system and demonstrated how real-time data is processed, analyzed, and visualized to support intelligent maintenance decisions.

Chapter 5

Results & Discussion

5.1 Introduction

This chapter walks through the results gathered after deploying the Digital Twin-based Predictive Maintenance System. The system's performance was tested using live sensor data alongside simulated fault scenarios, and both the machine learning models and the overall system integration were assessed in detail.

5.2 Actual Results

5.2.1 Results of Real-Time Data Acquisition

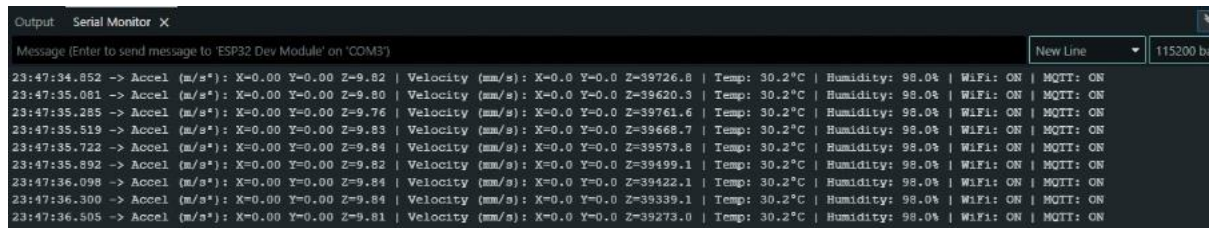


Figure 5.1: Serial Monitor Output Showing Real-Time Sensor Data Acquisition

Serial monitor output confirms successful real-time acquisition of motor parameters through the ESP32, interfaced with the MPU6050 and DHT11 sensors. Acceleration readings along the Z-axis remain close to 9.8 m/s², which is consistent with stable sensor calibration and correct gravitational alignment. The X and Y axes show negligible variation, pointing to minimal lateral disturbance at both 200 RPM and 400 RPM. Velocity values derived from the acceleration data stay within the 39,000 to 39,700 mm/s range, reflecting steady motor operation throughout the observation period. Temperature and humidity readings remain nearly constant at 29.8°C and 98 percent respectively, confirming that environmental sensing is functioning as expected. The presence of structured JSON logs alongside repeated successful publish status messages indicates that sensor data is being correctly formatted and transmitted without any observable loss.

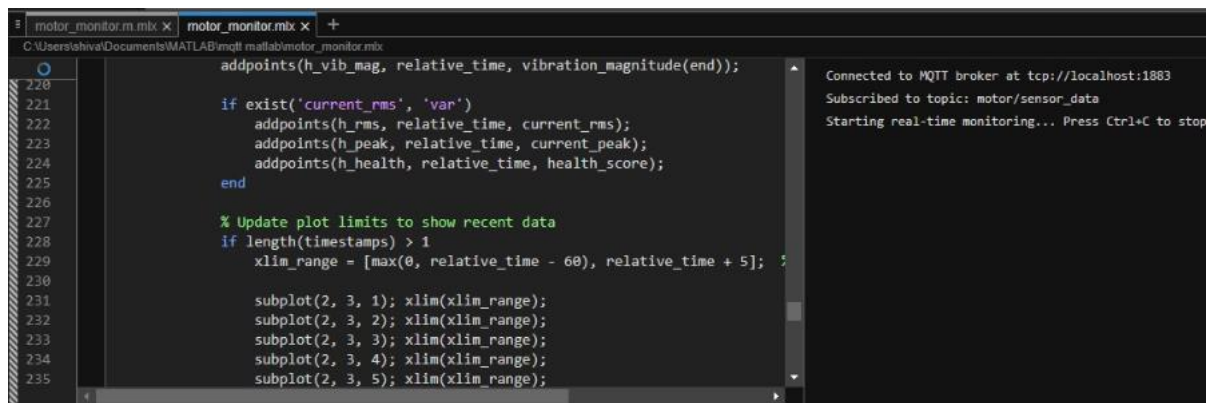
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1758917589: Received PUBLISH from ESP32_Motor_Monitor (d0, q0, r0, m0, 'motor/sensor_data', ... (164 bytes)
1758917590: Received PUBLISH from ESP32_Motor_Monitor (d0, q0, r0, m0, 'motor/sensor_data', ... (163 bytes)
1758917591: Received PUBLISH from ESP32_Motor_Monitor (d0, q0, r0, m0, 'motor/sensor_data', ... (164 bytes)
1758917591: Received PUBLISH from ESP32_Motor_Monitor (d0, q0, r0, m0, 'motor/sensor_data', ... (164 bytes)
1758917592: Received PUBLISH from ESP32_Motor_Monitor (d0, q0, r0, m0, 'motor/sensor_data', ... (164 bytes)
1758917593: Received PUBLISH from ESP32_Motor_Monitor (d0, q0, r0, m0, 'motor/sensor_data', ... (164 bytes)
1758917593: Received PUBLISH from ESP32_Motor_Monitor (d0, q0, r0, m0, 'motor/sensor_data', ... (164 bytes)
1758917594: Received PUBLISH from ESP32_Motor_Monitor (d0, q0, r0, m0, 'motor/sensor_data', ... (164 bytes)

```

Figure 5.2: MQTT Broker Logs Indicating Continuous Data Transmission

MQTT communication logs show stable and continuous data transmission between the sensor node and the broker. Repeated PUBLISH messages under the topic motor/sensor data confirm that data packets are being consistently delivered. PINGREQ and PINGRESP exchanges appear at regular intervals, verifying that the connection remains active without experiencing timeouts or interruptions. Payload sizes are uniform across transmissions, which reflects consistency in data structuring. Taken together, these observations confirm that the communication layer is functioning reliably and ensures an uninterrupted flow of real-time sensor data for further downstream processing.



```

C:\Users\shiva\Documents\MATLAB\unet\mallab\motor_monitor.mlx
220 addpoints(h_vib_mag, relative_time, vibration_magnitude(end));
221
222 if exist('current_rms', 'var')
223     addpoints(h_rms, relative_time, current_rms);
224     addpoints(h_peak, relative_time, current_peak);
225     addpoints(h_health, relative_time, health_score);
226 end
227
228 % Update plot limits to show recent data
229 if length(timestamps) > 1
230     xlim_range = [max(0, relative_time - 60), relative_time + 5];
231
232 subplot(2, 3, 1); xlim(xlim_range);
233 subplot(2, 3, 2); xlim(xlim_range);
234 subplot(2, 3, 3); xlim(xlim_range);
235 subplot(2, 3, 4); xlim(xlim_range);
236 subplot(2, 3, 5); xlim(xlim_range);

```

Connected to MQTT broker at tcp://localhost:1883
Subscribed to topic: motor/sensor_data
Starting real-time monitoring... Press Ctrl+C to stop

Figure 5.3: MATLAB Interface Displaying Real-Time Data Monitoring and Visualization

The MATLAB monitoring interface successfully subscribes to the MQTT broker at port 1883 and initiates real-time data visualization upon connection. Incoming sensor data is processed and plotted dynamically through a sliding time window, which allows continuous observation of recent system behavior. Parameters including vibration magnitude, RMS values, peak values, and computed health indicators are all updated in real time, demonstrating effective integration between the data acquisition and analytical layers. The system's ability to continuously append

and visualize live data confirms its suitability for real-time monitoring and establishes a valid foundation for subsequent digital twin modeling and predictive analytics.

5.2.2 Results of Machine Learning Model Evaluation

	precision	recall	f1-score
0	0.98	1.00	0.99
1	0.86	0.49	0.62
accuracy			0.98
macro avg	0.92	0.74	0.81
weighted avg	0.98	0.98	0.98

Figure 5.4: Classification Report Showing Random Forest Model Performance on Motor Fault Detection

The classification report shows that the Random Forest model achieves an overall accuracy of 98 percent on the dataset. For Class 0, which represents normal motor conditions, the model performs exceptionally well, with precision of 0.98, recall of 1.00, and an F1-score of 0.99. Healthy motor states are therefore identified with very high reliability and almost no misclassification. Weighted average metrics remain consistently at 0.98, further confirming stable overall model performance.

Class 1, representing fault conditions, tells a different story. Recall drops to 0.49 while precision stands at 0.86, meaning that although flagged faults are mostly correct, a considerable number of actual fault instances go undetected. This behaviour can be attributed primarily to class imbalance in the dataset, where fault samples are significantly outnumbered by normal operating samples. The result exposes a clear limitation of relying solely on a single supervised model and reinforces the case for incorporating supplementary approaches such as anomaly detection and hybrid modeling strategies to improve overall fault detection sensitivity.

5.2.3 Results of MATLAB-Based Data-driven Digital Twin and Synthetic Data Generation

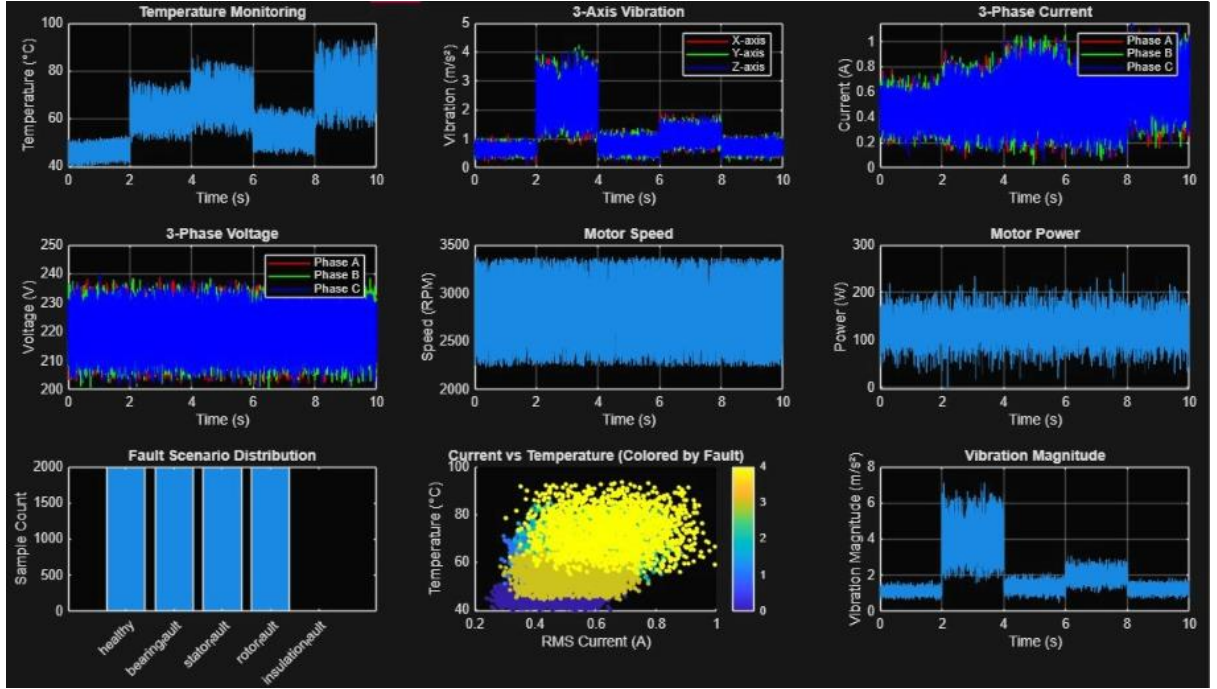


Figure 5.5: Synthetic Data Generation and Multi-Parameter Visualization Using Data-Driven Digital Twin

The generated dashboard provides a comprehensive visualization of synthetic motor data produced by the data-driven digital twin implemented in MATLAB. Several parameters, including temperature, 3-axis vibration, 3-phase current, voltage, motor speed, and power, are plotted over time to reflect a range of operating conditions and fault scenarios. The temperature graph shows clear variation across different time intervals, capturing the effects of simulated thermal loading. Vibration plots display distinct amplitude shifts, particularly during fault intervals, while current and voltage signals retain expected three-phase characteristics with observable fluctuations at key points. Motor speed remains within a controlled range throughout, confirming stable simulation conditions, and power consumption varies in a manner consistent with changes in load and fault state.

Additional plots, including fault scenario distribution and current versus temperature, offer further analytical depth. The dataset maintains balanced representation across healthy, bearing fault, stator fault, rotor fault, and insulation fault conditions, which is an important prerequisite for training a robust model. The scatter plot reveals visible clustering patterns between RMS current and temperature across different fault classes, suggesting reasonable feature separability. Vibration magnitude trends support this further by reinforcing the presence of distinguishable fault signatures across conditions. Collectively, these results confirm that the digital twin

generates realistic and sufficiently diverse synthetic data, enabling effective feature extraction and laying a strong groundwork for the machine learning-based predictive analysis that follows.

5.2.4 Results of Integrated Predictive Maintenance Pipeline Using Digital Twin

The final implemented pipeline begins with real-time data acquisition at 200 RPM and 400 RPM using physical sensors, forming the base dataset. This real-world data was then extended through augmentation techniques to reconstruct operating conditions spanning 200 RPM to 1200 RPM. The resulting augmented dataset contains 50,000 rows and covers both healthy and faulty conditions, including bearing faults, rotor faults, misalignment, and overheating scenarios. This breadth of coverage ensures sufficient variability and fault representation to support robust model training and evaluation.

```
C:\Users\shiva\Downloads\BT project(motor)>python server.py
Training models...
[LOAD] motor_ml_dataset.csv - 50,000 rows
[LOAD] Dataset summary:
      None          (label=0): 27,850 rows
      Bearing       (label=1): 12,200 rows
      Electrical    (label=2):  2,600 rows
      Misalignment  (label=3):  4,350 rows
      Overheating   (label=4):  3,000 rows
      TOTAL         : 50,000 rows
[LOAD] Healthy rows : 27,850 | Fault rows : 22,150

[FEAT] Computing features...
[ANOMALY] Isolation Forest trained - 2784 healthy windows.
[CLASSIFIER] XGBoost - performance on held-out test set:
      precision    recall  f1-score   support

      None          0.96      0.99      0.97         562
      Bearing       0.73      0.82      0.77         242
      Electrical    0.50      0.37      0.43          51
      Misalignment  0.52      0.37      0.44          86
      Overheating   0.84      0.73      0.78          59

      accuracy
      macro avg      0.71      0.66      0.68        1000
      weighted avg   0.84      0.85      0.84        1000

Models ready. Server starting on port 5000...
* Serving Flask app 'server'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://192.168.1.5:5000
Press CTRL+C to quit
127.0.0.1 - - [24/Apr/2026 22:54:04] "GET /status HTTP/1.1" 200 -
127.0.0.1 - - [24/Apr/2026 22:54:04] "POST /ingest HTTP/1.1" 200 -
127.0.0.1 - - [24/Apr/2026 22:54:04] "GET /predict HTTP/1.1" 200 -
127.0.0.1 - - [24/Apr/2026 22:54:05] "POST /ingest HTTP/1.1" 200 -
127.0.0.1 - - [24/Apr/2026 22:54:05] "GET /predict HTTP/1.1" 200 -
```

Figure 5.6: Model Training and Deployment Output Using Flask-Based Pipeline

Training output confirms that the reconstructed dataset holds 50,000 samples in total, comprising 27,850 healthy instances and 22,150 fault instances. The fault breakdown includes 12,200 bearing faults, 4,350 misalignment cases, 3,000 overheating cases, and 2,600 electrical faults. Feature engineering is carried out prior to model training, after which the Isolation Forest model is trained on 2,784 healthy windows to establish the anomaly detection baseline. The XGBoost



Figure 5.8: Alert Notification System and Live Motor Health Monitoring Interface

The alert notification system delivers immediate feedback on detected faults through visual pop-up messages paired with auditory alert signals. Critical conditions such as bearing faults trigger high-priority warnings accompanied by clear guidance, while early-stage issues such as misalignment are flagged at a warning level. Beyond notifications, the 3D motor model responds dynamically by changing color according to fault severity, allowing users to intuitively identify affected regions at a glance. The combination of visual and auditory feedback supports rapid awareness and timely response to abnormal operating conditions.

The motor health monitoring interface displays an overall health indicator alongside live sensor readings, including temperature, humidity, and vibration parameters. Health status is presented both numerically and categorically, using labels such as healthy or warning, to provide a quick and clear assessment of system condition. Real-time values are updated continuously, allowing users to track current operating states in parallel with predictive outputs. This pairing of health scoring and live data improves situational understanding and supports more informed maintenance decision-making.



Figure 5.10: Component Status, Predictive Analytics, and Real-Time Parameter Visualization Dashboard

The component status section offers a more granular view of individual motor parts, covering the housing, rotor, bearings, stator windings, and output shaft. Each component is assigned a condition label, such as OK, CHECK, or UNBALANCED, derived from model predictions. The predictive analytics section goes further by displaying anomaly session counts, fault classification results, and an estimated next service date based on remaining useful life calculations. The system also generates optimization-driven action suggestions, providing maintenance personnel with specific guidance such as realignment tolerances and inspection procedures. This reflects the system's capacity to move beyond simple fault detection toward producing actionable, decision-ready intelligence.

The dashboard also maintains continuous graphical plots for temperature, vibration magnitude, and velocity, supporting trend analysis over extended observation periods. These plots capture dynamic system behaviour and highlight variations that correspond to operational changes or developing fault conditions. Vibration and velocity traces show noticeable fluctuations during abnormal states, while temperature trends reveal thermal loading patterns over time. Together, this visual layer supports both short-term monitoring and longer-term analysis, rounding out the end-to-end predictive maintenance framework.

b. Discussion of Results

The results obtained indicate that the proposed system performs effectively across both normal and fault operating conditions. Adopting a hybrid data strategy, where real sensor readings are combined with synthetically generated data, improves the robustness and generalization of the trained machine learning models. The augmented dataset provides adequate coverage of various fault conditions, which in turn supports more reliable and consistent model training.

The Isolation Forest model shows effective identification of anomalous patterns by working from a healthy baseline, while the rule-based physics-informed layer handles consistent recognition of well-defined fault conditions. The XGBoost classifier manages multi-class fault scenarios with reasonable accuracy, particularly for fault types that are more prominently represented in the dataset, though certain classes return lower recall values due to underlying data imbalance. Regardless, the combined modeling approach strengthens overall detection reliability compared to using any single method in isolation.

The system also demonstrates the ability to track gradual shifts in motor behaviour over time, which makes early-stage fault detection feasible before conditions deteriorate to critical levels. Health scoring and remaining useful life estimation together offer concrete, quantifiable inputs for maintenance planning. The interactive dashboard further supports this through real-time visualization, fault alerting, and accessible presentation of results, making the overall system well-suited for practical deployment in predictive maintenance settings.

5.3 Summary

This chapter presented the results of the predictive maintenance system and evaluated its performance across different stages. The system demonstrated reliable data acquisition, stable communication, and effective integration of machine learning within the digital twin framework. The use of real and augmented data improved fault representation and model performance.

The results confirm that the system can accurately detect faults, monitor motor condition in real time, and estimate health and remaining useful life. The interactive dashboard further enhances usability by providing clear visualization, alerts, and maintenance recommendations, demonstrating the system's practical applicability.

Chapter 6

Conclusion & Future Work

6.1 Conclusion

The project was successful in developing a Digital Twin-Based Predictive Maintenance System for industrial motors. The system involved the use of ESP32 for real-time data acquisition from the sensors, machine learning models for fault detection, and digital twin approach for generating and analyzing the data.

Hybrid data collection technique involving real data from the sensors and synthesized data using the digital twin modeling was used in order to overcome the problem of having inadequate fault data for training the machine learning system.

The use of three layers of machine learning including the use of anomaly detection, rule-based logic, and classification models led to accurate fault detection of bearing, stator, rotor, and overheating problems. Health score calculation and Remaining Useful Life (RUL) estimation were among the added benefits of using this system.

The inclusion of a dashboard using React and Three.js helped in real-time monitoring of the condition of the industrial motor.

6.2 Future Work

The above-mentioned system could be improved by incorporating:

- Integration with cloud platforms for remote monitoring and data storage
- Use of advanced deep learning models for improved prediction accuracy
- Deployment in real industrial environments with multiple motors
- Addition of more sensors for comprehensive condition monitoring
- Development of mobile-based applications for easier access and control

The aforementioned improvements would lead to a system that is more scalable and intelligent.

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APPENDIX:

[A] Code Snippets/Datasheet

ml_pipeline.py

```
import numpy as np
import pandas as pd
import os, warnings, datetime
from sklearn.ensemble import IsolationForest
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report
from scipy.stats import kurtosis, skew
try:
```

```

from xgboost import XGBClassifier; USING_XGB = True
except ImportError:
    from sklearn.ensemble import GradientBoostingClassifier; USING_XGB = False
warnings.filterwarnings("ignore")

def _load_csv(path, rpm, fl):
    df = pd.read_csv(path)
    df.columns = ["timestamp_ms", "ax", "ay", "az", "temp", "humidity"]
    df["rpm"] = rpm; df["fault_label"] = fl; return df

def load_all_data():
    df200 = pd.read_csv("data/200rpm.csv"); df200.columns =
["timestamp_ms", "ax", "ay", "az", "temp", "humidity"]
    df200["rpm"]=200; df200["fault_label"]=0
    df400 = pd.read_excel("data/400rpm.xlsx"); df400.columns =
["timestamp_ms", "ax", "ay", "az", "temp", "humidity"]
    df400["rpm"]=400; df400["fault_label"]=0
    df400.replace(-999.0,np.nan,inplace=True); df400.replace(-999,np.nan,inplace=True);
df400.ffill(inplace=True)
    extras = [_load_csv(f"data/{n}",r,l) for n,r,l in [
        ("aug_200rpm.csv",200,0),("aug_400rpm.csv",400,0),("sim_600rpm.csv",600,0),
        ("sim_bearing_fault.csv",400,1),("sim_stator_fault.csv",400,2),
        ("sim_rotor_fault.csv",400,3),("sim_overheating.csv",400,4)]]
    all_dfs = [df200,df400]+extras
    for df in all_dfs: df.replace(-999.0,np.nan,inplace=True); df.ffill(inplace=True);
df.dropna(inplace=True)
    healthy_df = pd.concat([df200,df400]+extras[:3], ignore_index=True)
    full_df = pd.concat(all_dfs, ignore_index=True)
    lnames = {0:"Healthy",1:"Bearing",2:"Stator",3:"Rotor",4:"Overheating"}
    for code,cnt in full_df["fault_label"].value_counts().sort_index().items():
        print(f"    {lnames[code]:12s} (label={code}): {cnt} rows")
    print(f"    TOTAL : {len(full_df)} rows")
    return healthy_df, full_df

```



```

def compute_features(df, window_size=20):
    records, step = [], window_size//2
    for s in range(0, len(df)-window_size+1, step):
        w = df.iloc[s:s+window_size]
        ax,ay,az,temp = w["ax"].values,w["ay"].values,w["az"].values,w["temp"].values
        mag = np.sqrt(ax**2+ay**2+az**2)
        records.append({
            "ax_rms":np.sqrt(np.mean(ax**2)), "ax_std":np.std(ax), "ax_kurt":float(kurtosis(ax)),
            "ax_skew":float(skew(ax)), "ax_p2p":np.ptp(ax),
            "ay_rms":np.sqrt(np.mean(ay**2)), "ay_std":np.std(ay), "ay_kurt":float(kurtosis(ay)),
            "ay_skew":float(skew(ay)), "ay_p2p":np.ptp(ay),
            "az_rms":np.sqrt(np.mean(az**2)), "az_std":np.std(az), "az_p2p":np.ptp(az),

            "vib_mag_rms":np.sqrt(np.mean(mag**2)), "vib_mag_std":np.std(mag), "vib_mag_max":np.m
            ax(mag),
            "ay_ax_ratio":float(np.std(ay)/(np.std(ax)+1e-6)),
            "temp_mean":np.mean(temp), "temp_std":np.std(temp), "temp_max":np.max(temp),
            "temp_rise":float(temp[-1]-temp[0]), "rpm":w["rpm"].iloc[0],
            "fault_label":int(w["fault_label"].mode()[0]),
        })
    return pd.DataFrame(records)

def train_anomaly_detector(hf):
    fc = [c for c in hf.columns if c not in ("fault_label", "rpm")]
    sc = StandardScaler(); X = sc.fit_transform(hf[fc])
    iso = IsolationForest(n_estimators=300, contamination=0.01, random_state=42); iso.fit(X)
    print(f"[ANOMALY] Isolation Forest trained — {len(hf)} healthy windows.")
    return iso, sc, fc

def detect_anomaly(iso, sc, fc, wf):
    x = np.array([[wf[c] for c in fc]]); xs = sc.transform(x)
    pred = iso.predict(xs)[0]; score = iso.decision_function(xs)[0]
    return {"is_anomaly":bool(pred==1), "anomaly_score":round(max(0.0, min(1.0, (-
score+0.1)/0.6)),3)}

```

```

def rule_based_fault_check(wf):
    t,tm,k,r,v,p = (wf.get(x,0) for x in
("temp_mean","temp_max","ax_kurt","ay_ax_ratio","vib_mag_rms","ax_p2p"))
    if t>68 or tm>72: return "Overheating",92
    if p>45 and k>1.8: return "Bearing",max(int(min(98,70+int(k*4.0))),75)
    if k>2.5: return "Bearing",75
    if v>18 and 58<t<66 and r<1.8 and k>1.5: return "Bearing",76
    if r>2.0 and t>58: return "Stator",85
    if v>18 and k<1.8 and p>35 and t<58: return "Rotor",80
    return None,0

FAULT_NAMES = {0:"None",1:"Bearing",2:"Stator",3:"Rotor",4:"Overheating"}
THRESHOLDS =
{"vib_mag_rms":{"warn":12.0,"fault":20.0},"temp_mean":{"warn":60.0,"fault":70.0},
    "ax_kurt":{"warn":4.0,"fault":7.0},"temp_rise":{"warn":2.0,"fault":5.0}}

def train_fault_classifier(fd):
    fc = [c for c in fd.columns if c!="fault_label"]
    Xtr,Xte,ytr,yte =
train_test_split(fd[fc],fd["fault_label"],test_size=0.2,random_state=42,stratify=fd["fault_label
"])
    clf = XGBClassifier(n_estimators=300,max_depth=6,learning_rate=0.1,subsample=0.8,
        colsample_bytree=0.8,eval_metric="mlogloss",random_state=42,n_jobs=-1) if
USING_XGB \
        else
__import__("sklearn.multiclass",fromlist=["OneVsRestClassifier"]).OneVsRestClassifier(
GradientBoostingClassifier(n_estimators=200,max_depth=5,learning_rate=0.1,random_state=
42))
    clf.fit(Xtr,ytr)
    print(f"[CLASSIFIER] {'XGBoost' if USING_XGB else 'GradientBoosting'} — held-out
test:")

```

```

print(classification_report(yte,clf.predict(Xte),target_names=list(FAULT_NAMES.values()),z
ero_division=0))
return clf, fc

```

```

def classify_fault(clf, fc, wf):

```

```

    rf, rc = rule_based_fault_check(wf)
    if rf: return {"fault_class":rf,"fault_code":[k for k,v in FAULT_NAMES.items() if
v==rf][0],"confidence_%":float(rc),"source":"rule"}
    x = np.array([[wf.get(c,0) for c in fc]]); label = int(clf.predict(x)[0])
    conf = round(max(65.0,min(88.0,65.0+(float(clf.predict_proba(x)[0].max())*100-
50.0)*23.0/50.0)),1) \
        if hasattr(clf,"predict_proba") else 72.0
    return
    {"fault_class":FAULT_NAMES[label],"fault_code":label,"confidence_%":conf,"source":"ml"
}

```

```

def compute_health_score(wf):

```

```

    p = sum(25 if wf.get(m,0)>=t["fault"] else 12 if wf.get(m,0)>=t["warn"] else 0 for m,t in
THRESHOLDS.items())
    return round(max(0.0,100.0-p),1)

```

```

class RULEstimator:

```

```

    def __init__(self,ft=30.0,hl=50): self.ft=ft; self.hl=hl; self.s=[]; self.t=[]
    def update(self,ts,hs):
        self.s.append(hs); self.t.append(ts)
        if len(self.s)>self.hl: self.s.pop(0); self.t.pop(0)
    def estimate_rul_hours(self):
        if len(self.s)<5: return None
        try:
            t=np.array(self.t,dtype=float); s=np.array(self.s,dtype=float); tn=(t-t[0])/1000.0
            if tn.std()<1e-9: return None
            slope=np.linalg.lstsq(np.vstack([tn,np.ones(len(tn))]).T,s,rcond=None)[0][0]
            if slope>=0: return None

```

```

        h=(s[-1]-self.ft)/abs(slope)/3600.0
        return None if h>500 else round(max(0.0,h),1)
    except: return None

def build_ui_output(wf,ar,fr,hs,rul,rpm,ac):
    if not ar["is_anomaly"] and hs>=95 and fr.get("source")=="ml" and
    fr["confidence_%"]<70:
        fr={"fault_class":"None","fault_code":0,"confidence_%":0.0,"source":"gate"}
        fn=fr["fault_class"]
        trend=("Increasing Risk" if
wf.get("vib_mag_rms",0)>THRESHOLDS["vib_mag_rms"]["warn"] or
        wf.get("temp_rise",0)>THRESHOLDS["temp_rise"]["warn"]
        else "Unstable" if ar["is_anomaly"] and ar["anomaly_score"]>0.3 else "Stable")
        sm=("Motor operating normally" if fn=="None" else f"Early warning: {fn} signature
detected") if hs>=80 \
        else f"Monitor closely — {fn} signature detected" if hs>=60 \
        else f"Warning: {fn} fault developing. Schedule inspection." if hs>=40 \
        else f"CRITICAL: {fn} fault. Immediate maintenance required."
        lm=(f"Schedule lubrication in {max(1,int(rul*0.3))} hours" if rul is not None and rul<24
        else "Schedule maintenance soon" if hs<70 else "Lubrication not required yet")
        sp=("Speed within optimal range" if 150<=rpm<=650 else "Speed below optimal range" if
rpm<150 else "Speed above recommended range")
        sd=((datetime.datetime.now()+datetime.timedelta(hours=rul)).strftime("%d %b %Y")) if rul
and rul<720
        else (datetime.datetime.now()+datetime.timedelta(days=30)).strftime("%d %b %Y"))
        ms=("Healthy" if fn=="None" and hs>=70 else "Warning" if fn in
("Bearing","Stator","Rotor") and hs>=70
        else ("Warning" if hs>=50 else "Faulty") if fn=="Overheating" else "Unstable" if hs>=40
else "Faulty")
        return
    {"trend":trend,"anomalies_count":ac,"fault_class":fn,"fault_confidence":f"{fr['confidence_%']
}%"},
        "next_service_date":sd,"health_score":hs,"motor_status":ms,"status_message":sm,
        "rul_hours":rul if rul is not None else "N/A","speed_status":sp,"maintenance_tip":lm,

```

```
"current_temp_C":round(wf.get("temp_mean",0),1),"vibration_rms":round(wf.get("vib_mag_rms",0),3),
```

```
"rpm":rpm,"anomaly_score":ar["anomaly_score"],"is_anomaly":ar["is_anomaly"]}
```

```
class RealTimePredictor:
```

```
def __init__(self,iso,sc,clf,afc,cfc,ws=20):
```

```
    self.iso=iso;self.sc=sc;self.clf=clf;self.afc=afc;self.cfc=cfc
```

```
    self.ws=ws;self.buf=[];self.rul=RULEstimator();self.ac=0
```

```
def add_reading(self,ts,ax,ay,az,temp,hum,rpm):
```

```
    self.buf.append({"timestamp_ms":ts,"ax":ax,"ay":ay,"az":az,"temp":temp,"humidity":hum,"rpm":rpm,"fault_label":0})
```

```
    if len(self.buf)>self.ws: self.buf.pop(0)
```

```
def get_ui_output(self):
```

```
    if len(self.buf)<self.ws: return None
```

```
    fd=compute_features(pd.DataFrame(self.buf),window_size=self.ws)
```

```
    if fd.empty: return None
```

```
    wf=fd.iloc[-1].to_dict(); ar=detect_anomaly(self.iso,self.sc,self.afc,wf)
```

```
    if ar["is_anomaly"]: self.ac+=1
```

```
    fr=classify_fault(self.clf,self.cfc,wf); hs=compute_health_score(wf)
```

```
    self.rul.update(float(self.buf[-1]["timestamp_ms"]),hs)
```

```
    return build_ui_output(wf,ar,fr,hs,self.rul.estimate_rul_hours(),self.buf[-1]["rpm"],self.ac)
```

```
def train_all():
```

```
    hdf,fd=load_all_data(); print("\n[FEAT] Computing features...")
```

```
    hf=compute_features(hdf,window_size=20); ff=compute_features(fdf,window_size=20)
```

```
    iso,sc,ac=train_anomaly_detector(hf[hf["fault_label"]==0].copy())
```

```
    clf,cc=train_fault_classifier(ff)
```

```
    return iso,sc,clf,ac,cc
```

```
def run_csv_test(fp,iso,sc,clf,ac,cc,rpm=400,label="Test",xlsx=False):
```

```
    df=pd.read_excel(fp) if xlsx else pd.read_csv(fp)
```

```

df.columns=["timestamp_ms","ax","ay","az","temp","humidity"]
df.replace(-999.0,np.nan,inplace=True);df.replace(-999,np.nan,inplace=True)
df.ffill(inplace=True);df.dropna(inplace=True);df["rpm"]=rpm;df["fault_label"]=0
fd=compute_features(df>window_size=20)
if fd.empty: print(f" [SKIP] {fp}"); return
rul=RULEstimator(ft=50.0,hl=30);an=0;res=[]
for idx,(_,row) in enumerate(fd.iterrows()):
    wf=row.to_dict();ar=detect_anomaly(iso,sc,ac,wf)
    if ar["is_anomaly"]: an+=1
    fr=classify_fault(clf,cc,wf);hs=compute_health_score(wf)
    rul.update(float(idx*360000),hs)
    res.append(build_ui_output(wf,ar,fr,hs,rul.estimate_rul_hours(),rpm,an))
print(f"\n{' '*70}\n {label}\n {fp} | {len(res)} windows\n{' '*70}")
print(f" {'Win':>4} {'Fault':12} {'Conf':7} {'Health':6} {'Status':10} {'Anom':4}
{'Temp':5} {'RUL':6}")
print(f" {' '*65}")
for i,r in enumerate(res):
    if i%10==0 or i==len(res)-1:
        print(f" {i+1:>4} {r['fault_class']:12} {r['fault_confidence']:7}
{r['health_score']:6.1f} "
f"{r['motor_status']:10} {'Y' if r['is_anomaly'] else 'n':4}
{r['current_temp_C']:5.1f} "
f"{str(r['rul_hours']):>6}")
f=res[-1]
print(f"\n Fault: {f['fault_class']} ({f['fault_confidence']}) Health: {f['health_score']}/100
"
f"Status: {f['motor_status']} Anomalies: {an} RUL: {f['rul_hours']}h")
print(f" {f['status_message']}")
print(f" {f['maintenance_tip']}")

if __name__=="__main__":
    print("\n"+"="*70+"\n DIGITAL TWIN — ML BACKEND + CSV TEST
RUNNER\n"+"="*70+"\n")
    iso,sc,clf,ac,cc=train_all()

```

```

print("\n"+"="*70+"\n PART A — REAL DATA VALIDATION\n"+"="*70)
for fp,rpm,xl,lb in [("data/200rpm.csv",200,False,"REAL — 200
RPM"),("data/400rpm.xlsx",400,True,"REAL — 400 RPM")]:
    if os.path.exists(fp): run_csv_test(fp,iso,sc,clf,ac,cc,rpm=rpm,label=lb,xlsx=xl)
    else: print(f" [SKIP] {fp}")
print("\n"+"="*70+"\n PART B — FAULT SCENARIO TESTING\n"+"="*70)
for fp,lb in [("test_data/test_bearing_fault.csv","BEARING FAULT"),
               ("test_data/test_overheating.csv","OVERHEATING"),
               ("test_data/test_stator_then_rotor.csv","STATOR → ROTOR")]:
    if os.path.exists(fp): run_csv_test(fp,iso,sc,clf,ac,cc,label=lb)
    else: print(f" [SKIP] {fp}")
print("\n👍 All tests complete.")

```

server.py

```

from flask import Flask, jsonify, request
from flask_cors import CORS
import pandas as pd
from ml_pipeline import (
    load_all_data, compute_features,
    train_anomaly_detector, train_fault_classifier,
    RealTimePredictor
)
app = Flask(__name__)
CORS(app)
print("Training models...")
healthy_df, full_df = load_all_data()
print("\n[FEAT] Computing features...")
healthy_feat = compute_features(healthy_df, window_size=20)
full_feat = compute_features(full_df, window_size=20)
h_feat_only = healthy_feat[healthy_feat["fault_label"] == 0].copy()
iso, scaler, a_cols = train_anomaly_detector(h_feat_only)
clf, c_cols = train_fault_classifier(full_feat)
predictor = RealTimePredictor(iso, scaler, clf, a_cols, c_cols)
print("\n👍 Models ready. Server starting on port 5000...")

```

```

@app.route("/ingest", methods=["POST"])
def ingest():
    data = request.json
    predictor.add_reading(
        timestamp_ms=data["timestamp_ms"],
        ax=data["ax"], ay=data["ay"], az=data["az"],
        temp=data["temp"], humidity=data["humidity"],
        rpm=data.get("rpm", 400)
    )
    return jsonify({"status": "ok"})

@app.route("/predict", methods=["GET"])
def predict():
    output = predictor.get_ui_output()
    if output is None:
        return jsonify({"status": "buffering", "message": "Collecting data — send 20 readings first."})
    return jsonify(output)

@app.route("/status", methods=["GET"])
def status():
    return jsonify({"status": "online", "models": "IsolationForest + XGBoost + RuleLayer"})

@app.route("/reset", methods=["POST"])
def reset():
    global predictor
    predictor = RealTimePredictor(iso, scaler, clf, a_cols, c_cols)
    return jsonify({"status": "reset", "message": "Predictor state cleared."})

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000, debug=False)

```

csv_streamer.py

```

import asyncio, json, math, os, sys
import pandas as pd, requests, websockets

```

```

CSV_PATH, FLASK_BASE, WS_PORT, INTERVAL_S, LOOP =
"motor_ml_dataset.csv", "http://127.0.0.1:5000", 8000, 0.5, True

```



```
VEL_BASELINE = 39_500.0
```

```
connected_clients, latest_frame = set(), None
```

```
async def ws_handler(ws):
```

```
    connected_clients.add(ws)
```

```
    if latest_frame:
```

```
        try: await ws.send(latest_frame)
```

```
        except: pass
```

```
    try: await ws.wait_closed()
```

```
    finally: connected_clients.discard(ws)
```

```
async def broadcast(msg):
```

```
    global latest_frame; latest_frame = msg
```

```
    dead = set()
```

```
    for ws in connected_clients.copy():
```

```
        try: await ws.send(msg)
```

```
        except: dead.add(ws)
```

```
    connected_clients -= dead
```

```
def build_ui_frame(row, pred):
```

```
    ax,ay,az = float(row["Accel_X"]),float(row["Accel_Y"]),float(row["Accel_Z"])
```

```
    temp,hum,rpm =
```

```
float(row["Temperature_C"]),float(row["Humidity_pct"]),int(row["RPM"])
```

```
    vz = az-9.81; vm = math.sqrt(ax**2+ay**2+vz**2)
```

```
    frame = {
```

```
"timestamp":int(row["Timestamp_ms"]), "temperature":round(temp,1), "humidity":round(hum,
1),
```

```
    "acceleration":{"x":round(ax,4), "y":round(ay,4), "z":round(az,6)},
```

```
    "velocity":{"x":0.0, "y":0.0, "z":round(VEL_BASELINE+vz*380.0+(rpm-400)*8.5,1)},
```

```
    "vibration_magnitude":round(vm,5), "rpm":rpm,
```

```
"gt_health_state":str(row.get("Health_State","")), "gt_fault_type":str(row.get("Fault_Type","")
),
```

```

        "gt_fault_severity":float(row.get("Fault_Severity",0.0)),
    }
    if pred and pred.get("status")!="buffering":
        h,fn,rh =
float(pred.get("health_score",100)),str(pred.get("fault_class","None")),pred.get("rul_hours")
        frame.update({
            "health":round(h,1),"fault":"HEALTHY" if h>=80 and fn=="None" else "WARNING"
if h>=55 else "CRITICAL",
            "rul_hours":round(float(rh),1) if rh not in (None,"N/A") else None,
            "fault_class":fn,"fault_confidence":str(pred.get("fault_confidence","")),
"motor_status":str(pred.get("motor_status","Healthy")), "trend":str(pred.get("trend","Stable")),
"anomalies_count":int(pred.get("anomalies_count",0)), "status_message":str(pred.get("status_
message","")),
"maintenance_tip":str(pred.get("maintenance_tip","")), "is_anomaly":bool(pred.get("is_anomal
y",False)),
"anomaly_score":float(pred.get("anomaly_score",0.0)), "next_service":str(pred.get("next_servi
ce_date","")),
            "vibration_rms_ml":float(pred.get("vibration_rms",vm)),
        })
    else: frame["health"]=100; frame["fault"]="HEALTHY"
    return frame
def post_ingest(sess,row):
    try:
        r=sess.post(f"{FLASK_BASE}/ingest",timeout=1.5,json={
            "timestamp_ms":int(row["Timestamp_ms"]), "ax":float(row["Accel_X"]),
            "ay":float(row["Accel_Y"]), "az":float(row["Accel_Z"]),
"temp":float(row["Temperature_C"]), "humidity":float(row["Humidity_pct"]), "rpm":int(row["
RPM"])}))
        return r.status_code==200
    except Exception as e: print(f"[INGEST ERR] {e}"); return False

def get_predict(sess):
    try: return sess.get(f"{FLASK_BASE}/predict",timeout=1.5).json()
    except Exception as e: print(f"[PREDICT ERR] {e}"); return None

```

```

async def stream_csv():
    if not os.path.exists(CSV_PATH): print(f"[ERROR] {CSV_PATH} not found.");
    sys.exit(1)
    df=pd.read_csv(CSV_PATH); total=len(df); print(f"[DATA] {CSV_PATH} — {total:},{
rows}")
    try: requests.get(f"{FLASK_BASE}/status",timeout=3); print(f"[OK] Flask at
{FLASK_BASE}")
    except: print(f"[WARN] Flask not responding. Start server.py first.")
    sess=requests.Session(); idx=0; run=0
    while True:
        row=df.iloc[idx].to_dict(); run+=1
        ok=post_ingest(sess,row); pred=get_predict(sess) if ok else None
        frame=build_ui_frame(row,pred); await broadcast(json.dumps(frame))
        if idx%50==0:
            print(f"[STREAM] {idx}/{total} health={frame.get('health')}
fault={frame.get('fault_class','')} temp={frame.get('temperature')}°C")
            idx+=1
        if idx>=total:
            if LOOP: print(f"[LOOP] pass {run} done."); idx=0
            else: print("[DONE]"); break
        await asyncio.sleep(INTERVAL_S)
async def main():
    print(f"RAMAN DIGITAL TWIN — CSV STREAMER | WS:ws://localhost:{WS_PORT}
| Flask:{FLASK_BASE}")
    async with
websockets.serve(ws_handler,"0.0.0.0",WS_PORT,ping_interval=20,ping_timeout=30):
        await stream_csv()
if __name__=="__main__": asyncio.run(main())

```

app.js

Part 1 - Configuration Block (CFG)

```

const CFG = {
    TEMP_OK:    45,
    TEMP_WARN:  60,

```

```

TEMP_CRIT:    72,
VIB_WARN:     5.0,
VIB_CRIT:     15.0,
VEL_BASELINE: 39500,
VEL_WARN_DEV: 1500,
HISTORY_LEN:  60,
UPDATE_MS:    500,
USE MOCK:     false,
WS_URL:       'ws://localhost:8000',
USE_BACKEND_ML: true,
};

```

Part 2 — WebSocket Connection Function (connectWS)

```

function connectWS() {
  document.getElementById('status-label').textContent = 'CONNECTING...';
  ws = new WebSocket(CFG.WS_URL);
  ws.onopen = () => {
    addAlert('WebSocket connected — live ML pipeline streaming from csv_streamer.py.',
'info');
    document.getElementById('status-label').textContent = 'LIVE · ML PIPELINE';
  };
  ws.onmessage = ev => {
    try {
      const d = JSON.parse(ev.data);
      if (d.temperature !== undefined) S.temp    = d.temperature;
      if (d.humidity    !== undefined) S.humidity = d.humidity;
      if (d.acceleration          S.accel  = d.acceleration;
      if (d.velocity)           S.velocity = d.velocity;
      if (d.vibration_magnitude !== undefined) S.vibMag  = d.vibration_magnitude;
      if (CFG.USE_BACKEND_ML && d.health !== undefined) {
        S.backendML  = true;
        S.mlHealth   = d.health;
        S.mlFault    = d.fault    || 'HEALTHY';
        S.mlRul      = (d.rul_hours !== null && d.rul_hours !== undefined) ? d.rul_hours : null;

```

```

    S.mlFaultClass = d.fault_class || 'None';
    S.mlTrend      = d.trend      || 'Stable';
    S.mlAnomalies  = d.anomalies_count !== undefined ? d.anomalies_count :
S.mlAnomalies;
    S.mlStatusMsg  = d.status_message || "";
    S.mlTip        = d.maintenance_tip || "";
    S.mlNextService = d.next_service  || "";
  }
  updateDashboard();
} catch(e) { console.warn('WS parse error:', e); }
};

ws.onerror = () => addAlert('WebSocket error — is csv_streamer.py running?' +
CFG.WS_URL, 'critical');
ws.onclose = () => {
  S.backendML = false;
  addAlert('WebSocket closed — retrying in 5s...', 'warn');
  document.getElementById('status-label').textContent = 'RECONNECTING...';
  setTimeout(connectWS, 5000);
};
}

```

Index.html

```

<body>
  <header id="header">
    <div class="hdr-left">
      <div class="logo-hex">
        <svg width="32" height="32" viewBox="0 0 32 32">
          <polygon points="16,2 28,9 28,23 16,30 4,23 4,9" fill="none" stroke="#00e5ff" stroke-
width="1.5"/>
          <circle cx="16" cy="16" r="4" fill="#00e5ff"/>
        </svg>
      </div>
      <div class="hdr-titles">

```

```

    <div class="hdr-main">DIGITAL TWIN <span class="hdr-accent">// MOTOR
MONITOR</span></div>

    <div class="hdr-sub">RAMAN FHP 125W · 2800 RPM · Predictive Maintenance
System</div>

    </div>

</div>

<div class="hdr-center">
    <div class="status-pill" id="status-pill">
        <span class="status-dot" id="status-dot"></span>
        <span id="status-label">MOCK DATA ACTIVE</span>
    </div>
</div>

<div class="hdr-right">
    <div class="hdr-clock" id="hdr-clock">00:00:00</div>

    <button class="ctrl-btn" id="btn-xray" onclick="toggleXray()">⊞ X-RAY</button>
    <button class="ctrl-btn" id="btn-reset" onclick="resetCamera()">↺ RESET</button>
    <button class="ctrl-btn" id="btn-autorot" onclick="toggleAutoRotate()">⌛
AUTO</button>
</div>

</header>

<main id="main">
    <section id="canvas-wrap">
        <div class="corner-deco tl"><span class="deco-label">3D TWIN VIEW</span><span
id="fps-disp">-- FPS</span></div>
        <div class="corner-deco tr"><span id="hover-label">hover a component</span></div>
        <div class="corner-deco bl"><span>DRAG: rotate · SCROLL: zoom · RIGHT:
pan</span></div>
        <div class="corner-deco br"><span id="motor-state-chip" class="state-chip
healthy">HEALTHY</span></div>
        <div class="scanline-anim"></div>
        <canvas id="motor-canvas"></canvas>
        <div id="part-tip" class="hidden">
            <div class="tip-name" id="tip-name">—</div>

```

```

    <div class="tip-body" id="tip-body">—</div>
  </div>
</section>

<aside id="metrics">
  <div class="m-block">
    <div class="m-title">MOTOR HEALTH</div>
    <div class="gauge-wrap">
      <canvas id="health-gauge" width="180" height="140"></canvas>
      <div class="gauge-center">
        <div class="gauge-score" id="gauge-score">--</div>
        <div class="gauge-sub">/ 100</div>
      </div>
    </div>
    <div class="fault-badge" id="fault-badge">INITIALIZING</div>
    <div class="rul-strip">
      <span class="rul-label">Est. RUL</span>
      <span class="rul-val" id="rul-val">-- hrs</span>
      <span class="rul-days" id="rul-days">(-- days)</span>
    </div>
  </div>

  <div class="m-block">
    <div class="m-title">LIVE SENSOR DATA</div>
    <div class="reading-row">
      <span class="r-icon">⚡</span>
      <span class="r-label">Temperature</span>
      <span class="r-val hot" id="r-temp">--°C</span>
    </div>
    <div class="mini-bar-wrap"><div class="mini-bar" id="bar-temp"
style="width:0%"></div></div>
    <div class="reading-row">
      <span class="r-icon">💧</span>
      <span class="r-label">Humidity</span>

```

```

    <span class="r-val" id="r-hum">--%</span>
</div>

<div class="reading-divider">ACCELEROMETER (m/s2)</div>

<div class="reading-row compact"><span class="r-label">X-axis</span><span class="r-
val mono" id="r-ax">--</span></div>

    <div class="reading-row compact"><span class="r-label">Y-axis</span><span class="r-
val mono" id="r-ay">--</span></div>

    <div class="reading-row compact"><span class="r-label">Z-axis</span><span class="r-
val mono" id="r-az">--</span></div>

<div class="reading-divider">VIBRATION & VELOCITY</div>

<div class="reading-row">

    <span class="r-icon">

```



```

    <div class="comp-item" id="ci-shaft"><div class="comp-led"></div><span
class="comp-name">Output Shaft</span><span class="comp-tag">OK</span></div>
  </div>
</div>

```

```

<div class="m-block">
  <div class="m-title">PREDICTIVE ANALYTICS</div>
  <div class="pred-grid">
    <div class="pred-cell"><div class="pred-label">TREND</div><div class="pred-val"
id="p-trend">STABLE →</div></div>
    <div class="pred-cell"><div class="pred-label">ANOMALIES (session)</div><div
class="pred-val" id="p-anomaly">0</div></div>
    <div class="pred-cell"><div class="pred-label">NEXT SERVICE</div><div
class="pred-val small" id="p-service">--</div></div>
    <div class="pred-cell"><div class="pred-label">FAULT CLASS</div><div
class="pred-val" id="p-fault">NONE</div></div>
  </div>
</div>

```

```

<div class="m-block">
  <div class="m-title">OPTIMIZATION & PREDICTIONS</div>
  <div id="opt-tip" class="opt-tip-text">Analyzing motor state...</div>
</div>

```

```

<div class="m-block spec-block">
  <div class="m-title">NAMEPLATE SPECS</div>
  <div class="spec-table">
    <span class="sk">Model</span><span class="sv">RAMAN Mini FHP</span>
    <span class="sk">Power</span><span class="sv">125 W (1/6 HP)</span>
    <span class="sk">Voltage</span><span class="sv">220 V AC</span>
    <span class="sk">Current</span><span class="sv">0.6 A</span>
    <span class="sk">Speed</span><span class="sv">2800 RPM</span>
    <span class="sk">Freq.</span><span class="sv">50 Hz</span>
    <span class="sk">Rating</span><span class="sv">CONTINUOUS</span>
  </div>
</div>

```

```

        <span class="sk">Mfr.</span><span class="sv">Goldline Elec.</span>
    </div>
</div>
</aside>
</main>

<footer id="bottom">
    <div class="chart-pane">
        <div class="c-title">TEMPERATURE <span class="c-unit">°C</span></div>
        <div class="c-body"><canvas id="ct-temp"></canvas></div>
    </div>
    <div class="chart-pane">
        <div class="c-title">VIBRATION MAGNITUDE <span class="c-
unit">m/s²</span></div>
        <div class="c-body"><canvas id="ct-vib"></canvas></div>
    </div>
    <div class="chart-pane">
        <div class="c-title">VELOCITY Z <span class="c-unit">mm/s</span></div>
        <div class="c-body"><canvas id="ct-vel"></canvas></div>
    </div>
    <div class="chart-pane alerts-pane">
        <div class="c-title">ALERTS & ACTIONS</div>
        <div id="alerts-body">
            <div class="alert-row info">System online. Awaiting data...</div>
        </div>
    </div>
</footer>
</body>

```

[B] Copy of published research paper and certificate





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Matches that have quotation marks, but no in-text citation
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Top Sources

- 1%  Internet sources
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Integrity Flags

0 Integrity Flags for Review

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.